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# DEREK W EUBANKS, PE, HFDP, HBDP, CHD, CPD MISSION-CRITICAL & HEALTHCARE MECHANICAL ENGINEER | OMSCS (MACHINE LEARNING) – GEORGIA TECH GRADUATE STUDENT

## PROFESSIONAL SUMMARY

Licensed Mechanical Engineer (PE – CA, GA, TN) with 13 years of experience designing HVAC and hydronic infrastructure for mission critical facilities including hyperscale data centers, hospitals, and advanced manufacturing environments. Expert in Revit-based system design, chilled water plants, and high-performance ventilation systems. Currently pursuing a Master of Science in Computer Science (Machine Learning) at Georgia Tech as a Graduate Student through the OMSCS program - bringing a systems-level, data and machine learning–driven approach to engineering design and optimization.

## SKILLS & SOFTWARE

- Autodesk Revit Expert
- Bluebeam Revu
- Microsoft Office Suite
- Python, C# (Revit API), VBA
- D3.js, Matplotlib, Seaborn, Pandas, NumPy, SciPy, Scikit-learn
- Trane Trace 3D Plus; Carrier HAP
- High-performance Mission-Critical and Healthcare HVAC & Plumbing & Medical Gas Design
- Hydronic waterside plant design
- QA/QC for HVAC, plumbing, and medical gas drawing markups

## EXPERIENCE

**SENIOR MECHANICAL ENGINEER • SPACEX • JULY 2025 – JAN 2026; ON-SITE**
Led design and optimization of HVAC and hydronic systems supporting SpaceX advanced manufacturing facilities. Performed detailed load calculations in Trace 3D Plus and engineered variable-primary-flow heat pump chiller plants, coordinating full 3D Revit MEP models from field verification through system design. Delivered complete demolition and construction drawing packages—from laser-verified field data to permit-ready documentation—in <= one week.

**SENIOR ASSOCIATE | SENIOR MECHANICAL ENGINEER • SYSKA HENNESSEY GROUP • JAN 2024 – MARCH 2025; ON-SITE**
Led mechanical design for a 48 MW hyperscale Tier III data center, developing N+1 fault-tolerant HVAC and chilled-water infrastructure supporting data halls and critical electrical spaces. Modeled complex CRAH galleries, 40 chiller yards, and (3) mechanical rooms in Revit, producing coordinated construction documents for 50k CFM airside CRAHs and 21.8k GPM hydronic system.

**SENIOR MECHANICAL ENGINEER • JACOBS • OCT 2022 – DEC 2023; REMOTE**
Designed HVAC systems for Tesla and Rivian manufacturing facilities, modeling chilled-water AHU systems and performing load calculations using Trace 700. Modeled and coordinated multiple 40k+ CFM chilled-water AHUs and associated duct and hydronic systems within Revit to support large-scale manufacturing environments with low dewpoint requirements.

## EDUCATION

**BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING (BSME)**
**AUG 2007 - MAY 2013 • GEORGIA INSTITUTE OF TECHNOLOGY**
- Dean's List (2) semesters; Merit list (2) semesters
- Math GPA: 3.68/4.0 (7 courses); Computer Science GPA: 3.5/4.0 (2 courses)
- HOPE scholarship & Federal PELL Grant recipient for maintaining >= 3.0 GPA

**MASTER OF SCIENCE IN COMPUTER SCIENCE (MSCS)**
**AUG 2025 - DEC 2028 • GEORGIA INSTITUTE OF TECHNOLOGY**
- Specialization in Machine Learning
- Completed: CS 6603 (A), CSE 6242 (A)
- 4.0 GPA

*Full technical CV engineering portfolio available upon request or via GitHub.